Health × Stock
How the modules work together

Frappe Health and ERPNext Stock, explained as they behave on their own: what each module owns, how stock comes in and leaves, and exactly where care and stock meet.

Frappe, ERPNext and Frappe Health version 16 · every behaviour in this document was tried on a running site

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What this document covers

No custom screens, no extra code: only the two standard modules. Sequence diagrams show which record causes which. Where something surprised us, it is stated as a finding, with the test that showed it.

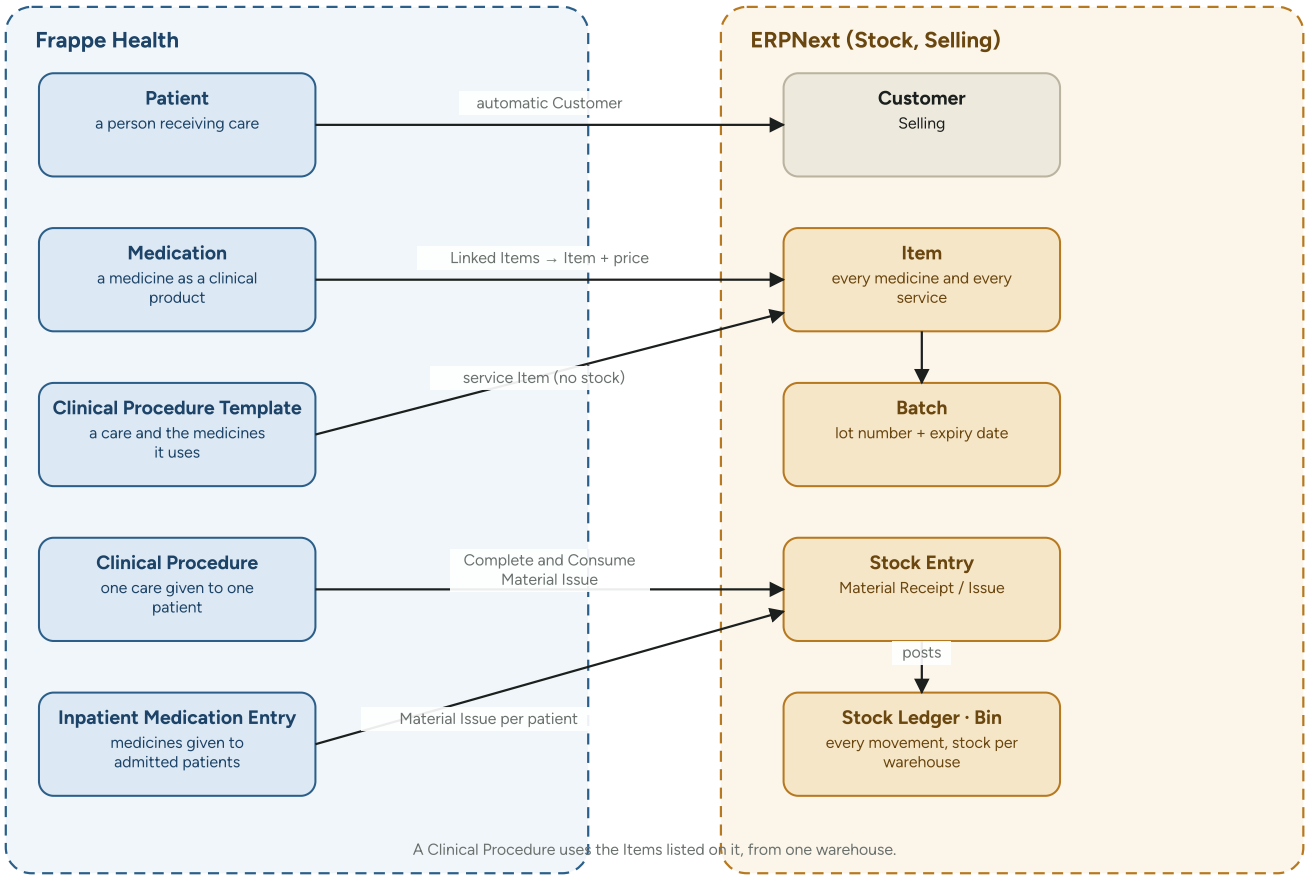
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Colours: person / other module · ERPNext Stock · Frappe Health · warning

OVERVIEW

The two modules at a glance

ERPNext Stock knows what is on the shelf. **Frappe Health** knows who received care. They meet at a small number of places, and almost all of them are the moment a procedure uses medicines.

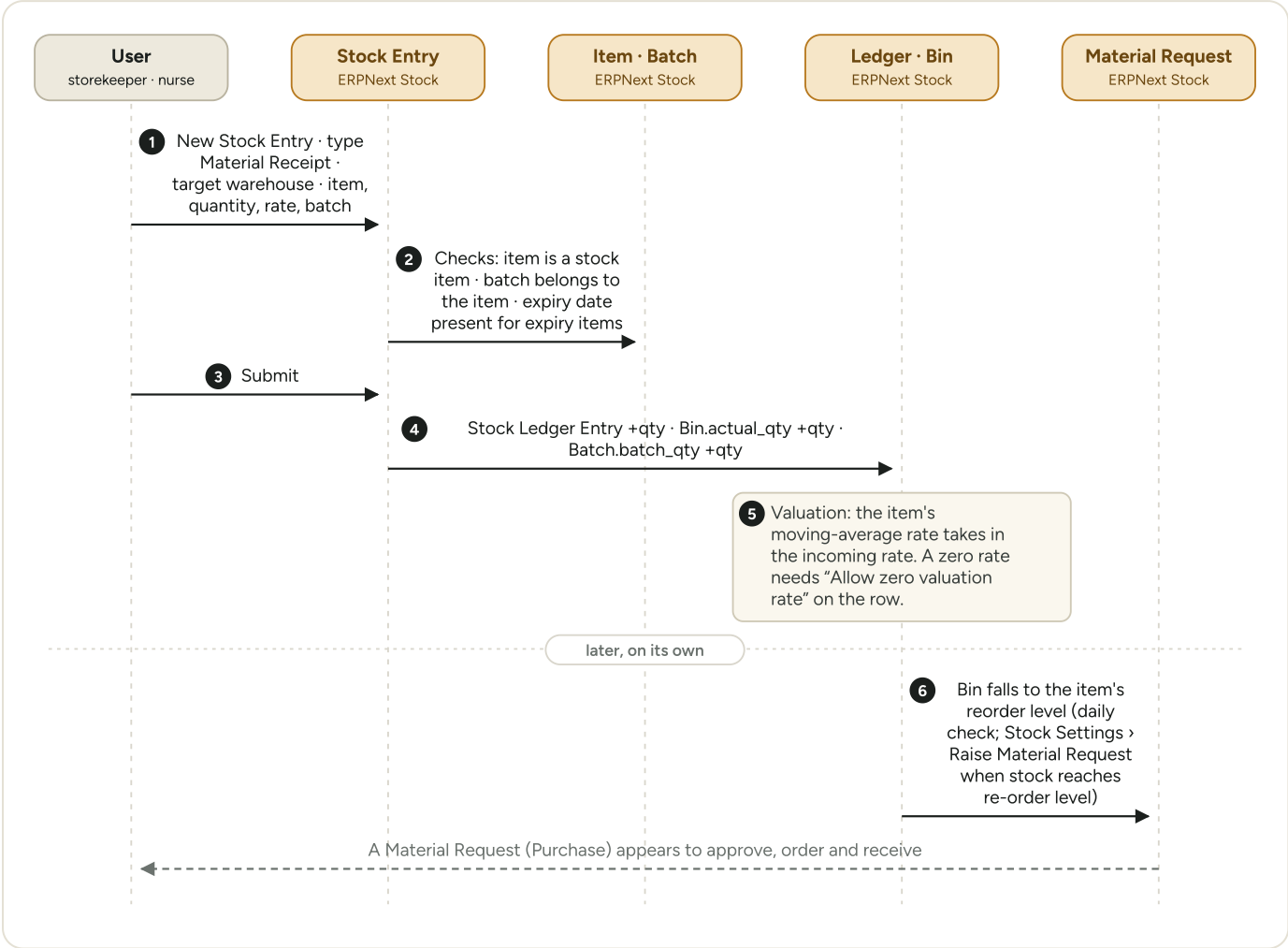


CONNECTION	TRIGGERED BY	WHAT IS CREATED
Medication → Item	Saving a Medication with linked items	An Item per linked row (disabled unless billable) and its Item Price
Patient → Customer	Saving a Patient	A Customer with the patient's name
Template → Item	Saving a procedure template	A service Item (no stock) and its price
Procedure → Stock Entry	Complete and Consume	A Material Issue from the procedure's warehouse
Inpatient Medication Entry → Stock Entry	Submit	A Material Issue with a row per patient order
Medication Request / prescription	Never	Nothing in stock: prescriptions only ask; billing is separate

STOCK MODULE

ERPNext Stock on its own

Every quantity lives in the Stock Ledger. A Stock Entry is the document that adds to it or takes from it. The Bin is the running total per item and warehouse; the Batch holds the same per lot.



STOCK IN: KEY FACTS

- Stock Entry type **Material Receipt** adds to a warehouse with no supplier. Purchases use Purchase Receipt (same ledger).
- Items with expiry need a Batch with an expiry date before they can be received.
- The batch quantity and the Bin always equal the sum of the ledger.

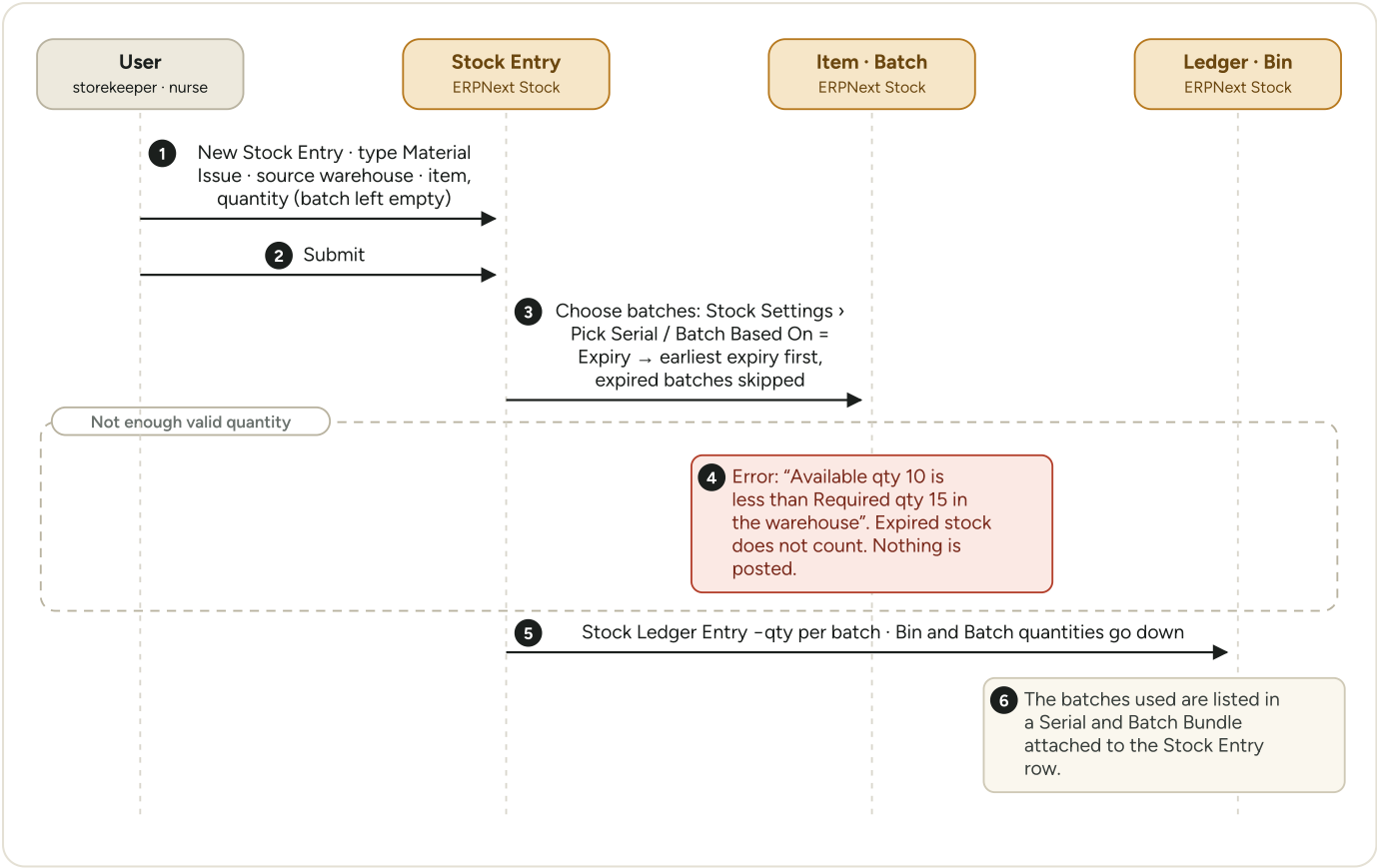
REORDER

- Each Item can have a reorder level per warehouse.
- With "Raise Material Request when stock reaches re-order level" on, a daily job raises a request when the Bin falls to that level.
- A Material Request never changes stock by itself.

STOCK MODULE

Stock out in ERPNext Stock

Stock leaves through a Stock Entry of type **Material Issue**. Batches are chosen by ERPNext, earliest expiry first, and expired lots are never used.



SETTINGS THAT MAKE THIS WORK

- Stock Settings › Pick Serial / Batch Based On = **Expiry**.
- Stock Settings › Use Serial / Batch fields = on (batch chosen on the row).
- Item › Has Batch No = yes, Has Expiry Date = yes.

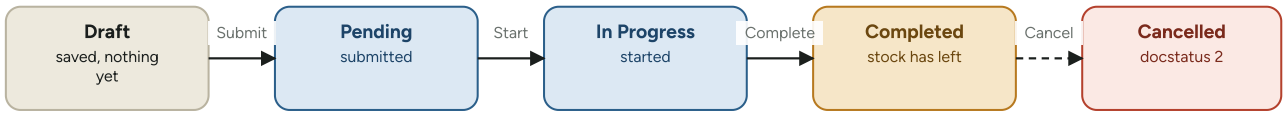
WHAT A USER SEES

- Material Issue with a source warehouse and the quantity.
- On submit: ledger rows per batch, the batches shown in the bundle.
- Too little valid stock: a clear error, nothing is posted.

HEALTH MODULE

Frappe Health on its own

Care is recorded on people and procedures. Only one document in the whole care flow, the Clinical Procedure, is able to move stock.



Start: checks the quantity in the warehouse. Short? ERPNext offers to record a Material Receipt for the shortage.
Complete and Consume: creates and submits the Stock Entry (Material Issue). This is the only moment stock moves.

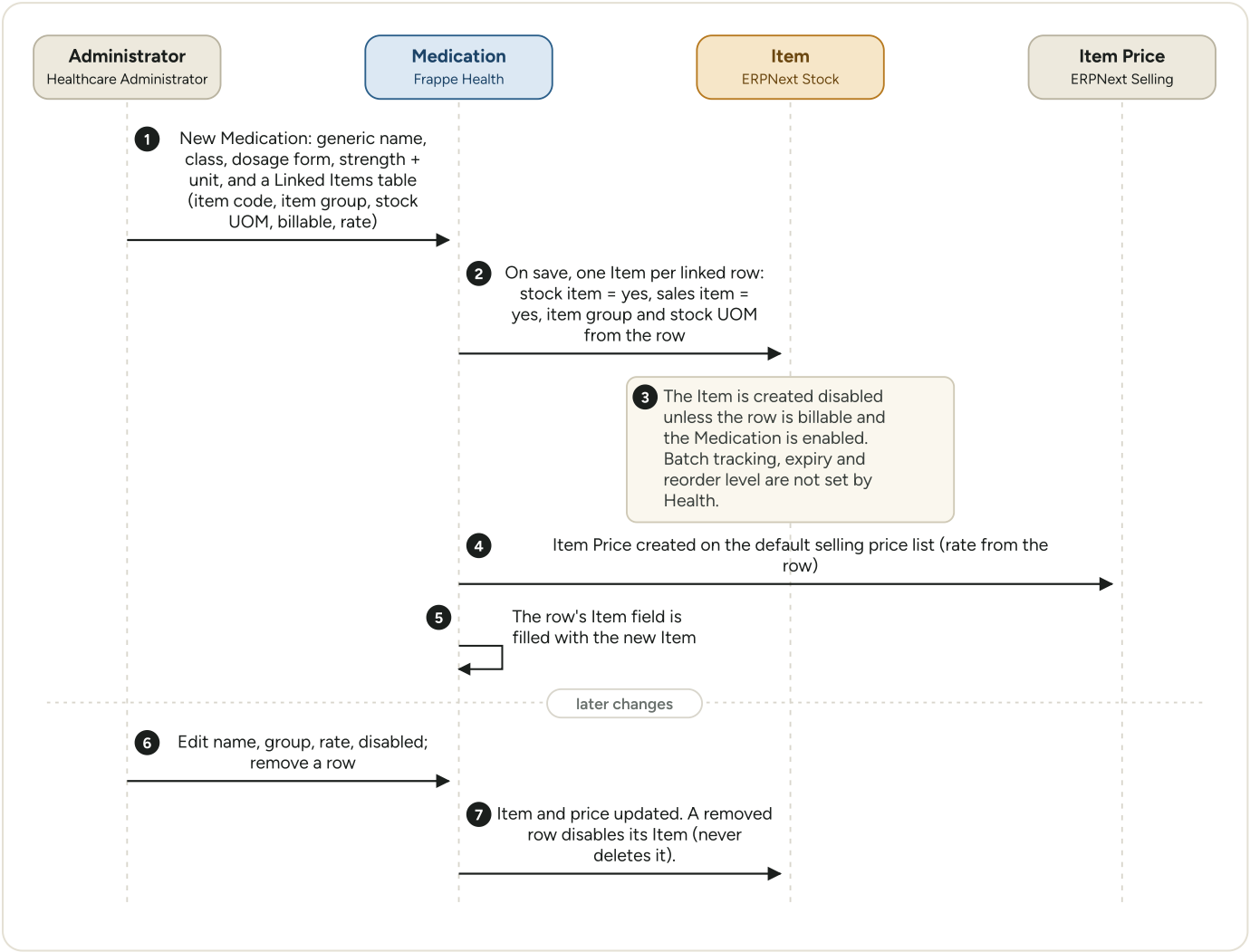
RECORD	HOLDS	MOVES STOCK?
Patient	Name, sex, birth date, phone, blood group, allergies, linked Customer	no
Medication	Generic name, class, dosage form, strength, linked Items	no
Clinical Procedure Template	Name, description, items + quantities (defaults)	no
Clinical Procedure	Patient, template, warehouse, date, items, status	**yes, when completed**
Inpatient Medication Order / Entry	Drug orders for admitted patients	yes, when the entry is submitted
Medication Request, Patient Encounter	Prescriptions and visits	no

Roles. Patients can be created by Nursing User, Physician and Laboratory User. Medications by Healthcare Administrator. Procedures by Nursing User and Physician. Details in “Permissions in the modules”.

CONNECTION 1

Medication and Item

A Medication is the clinical product; an Item is the stock product. Health creates the Item from the Medication's linked-items table.



FINDINGS (TESTED)

- The Item is created as a stock item and a sales item, disabled when the row is not billable.
- A Medication with no price list set needs the default selling price list, or it cannot be saved.
- The batch, expiry and reorder settings of an Item are not touched by Health.

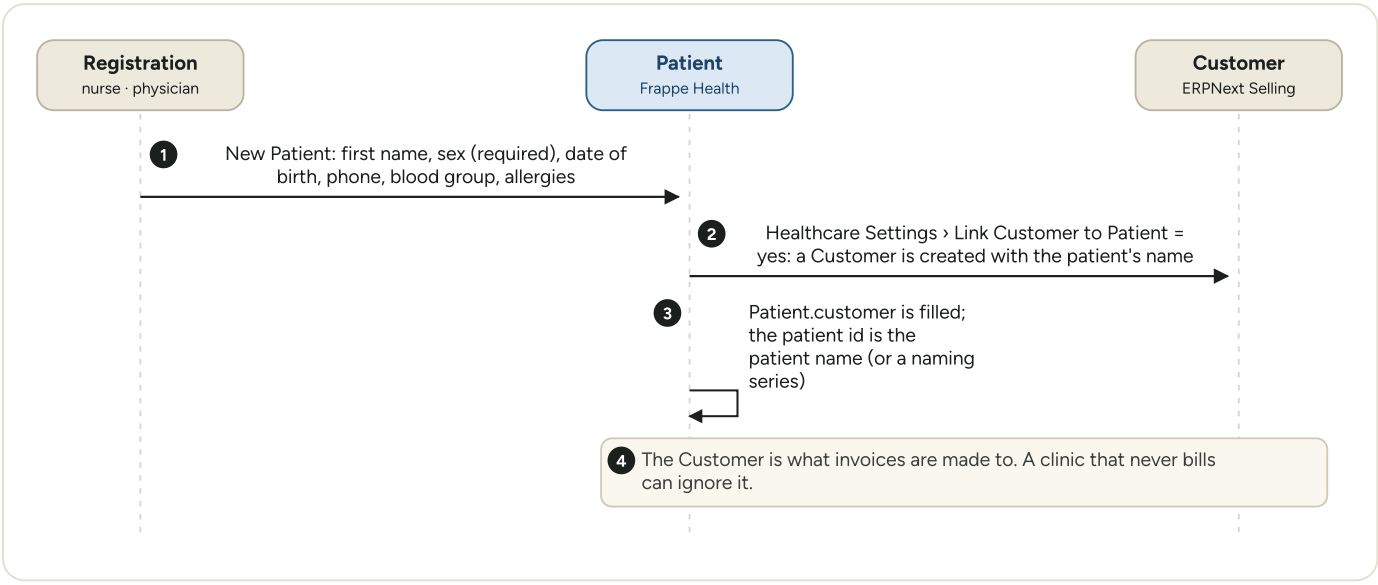
WHAT TO DO ABOUT IT

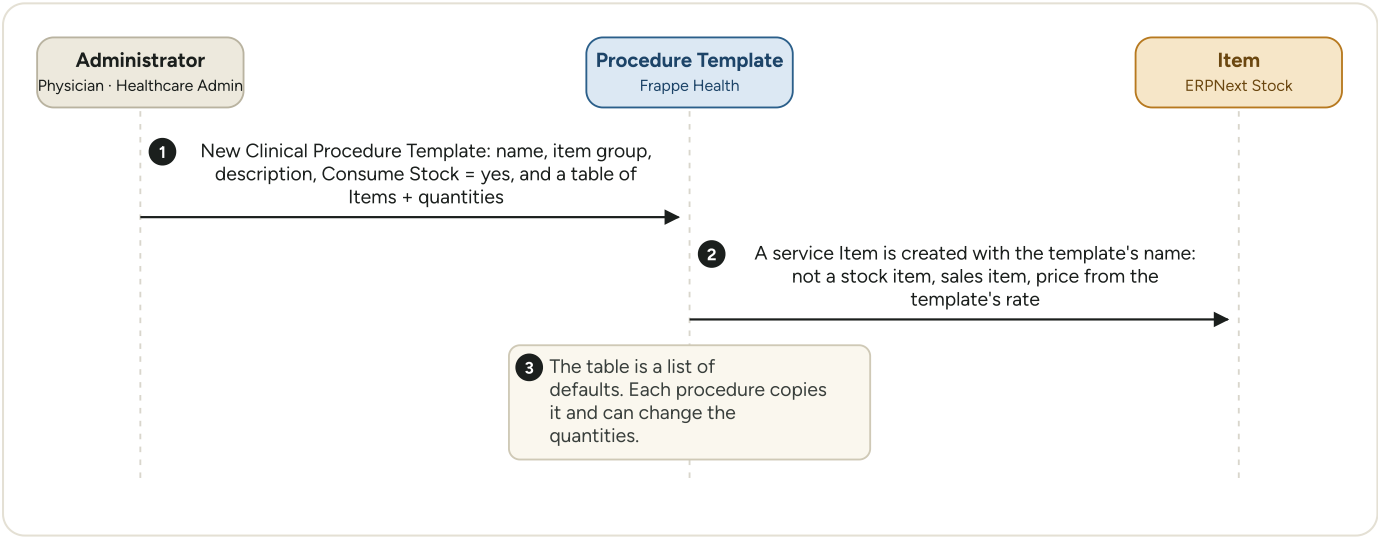
- Give the Item its batch, expiry and reorder settings in the Stock module (Item defaults can do this for every new Item).
- Enable the Item when the medicine is ready to use.
- Keep one Item per Medication: the Medication refuses an Item already linked elsewhere.

CONNECTIONS 2 AND 3

Patient and Customer, Template and Item

Two companions are created automatically. They matter for billing, not for stock.

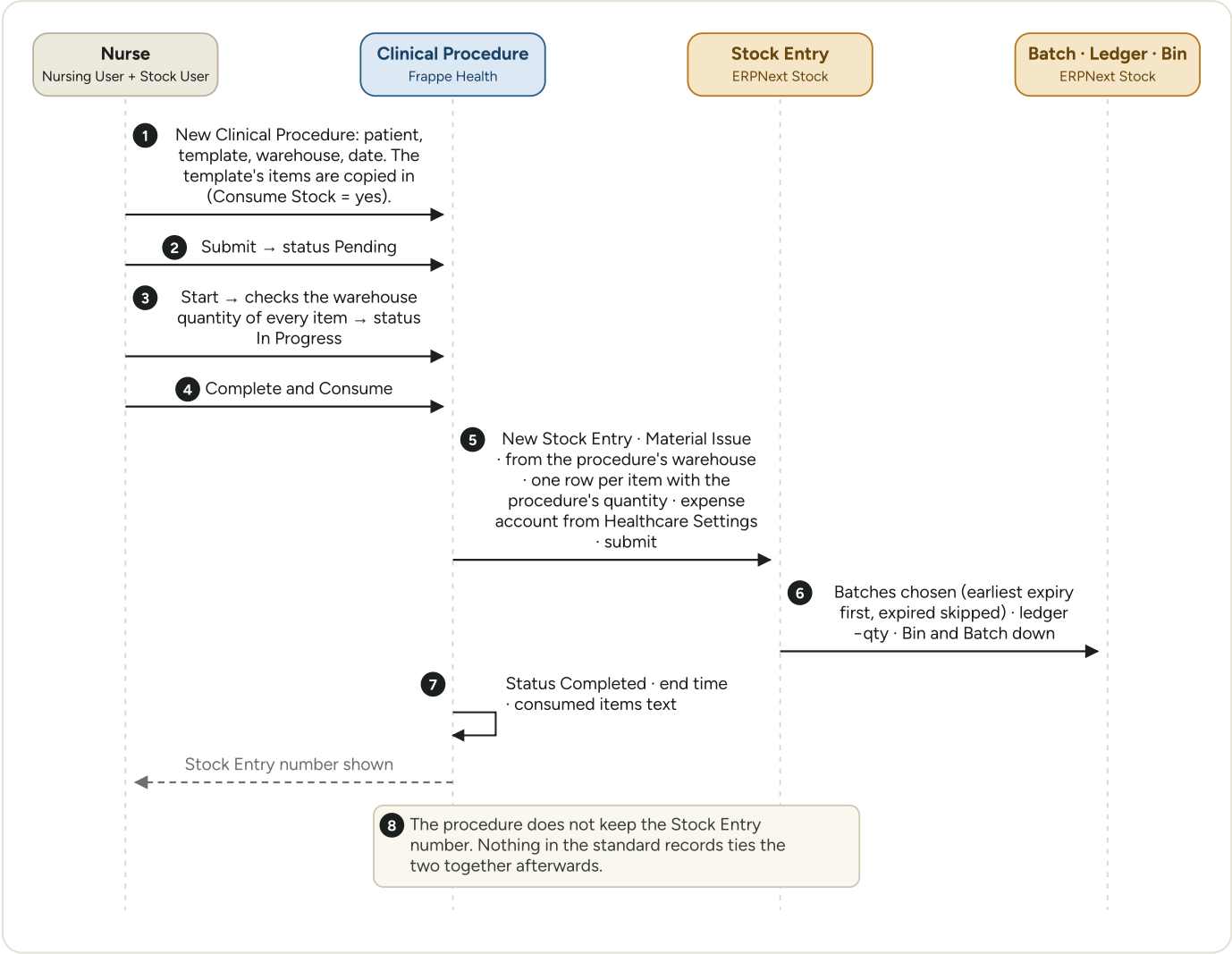




CONNECTION 4

Procedure to Stock Entry

This is the heart of the integration. A procedure lists the Items it uses. When it is completed, ERPNext issues them from the procedure's warehouse.



RULES (TESTED)

- Warehouse: the one chosen on the procedure. It is filled from the Healthcare Service Unit's warehouse when a service unit is picked; otherwise the nurse chooses it (Stock Settings has no default warehouse in version 16).
- Quantity is the procedure's own, not the template's: it can be changed per patient.
- Start checks the total quantity in the warehouse; if short it offers a Material Receipt for the shortage.

REMEMBER

- Standard cancel does not return stock (next pages).
- A procedure with Consume Stock = no never touches stock.
- The Stock Entry has no patient on it for procedures.

EXPERIMENTS

What ERPNext checks when stock leaves

Run on a site with Pick Serial / Batch Based On = Expiry. Item with two batches, procedures completed through the standard buttons.

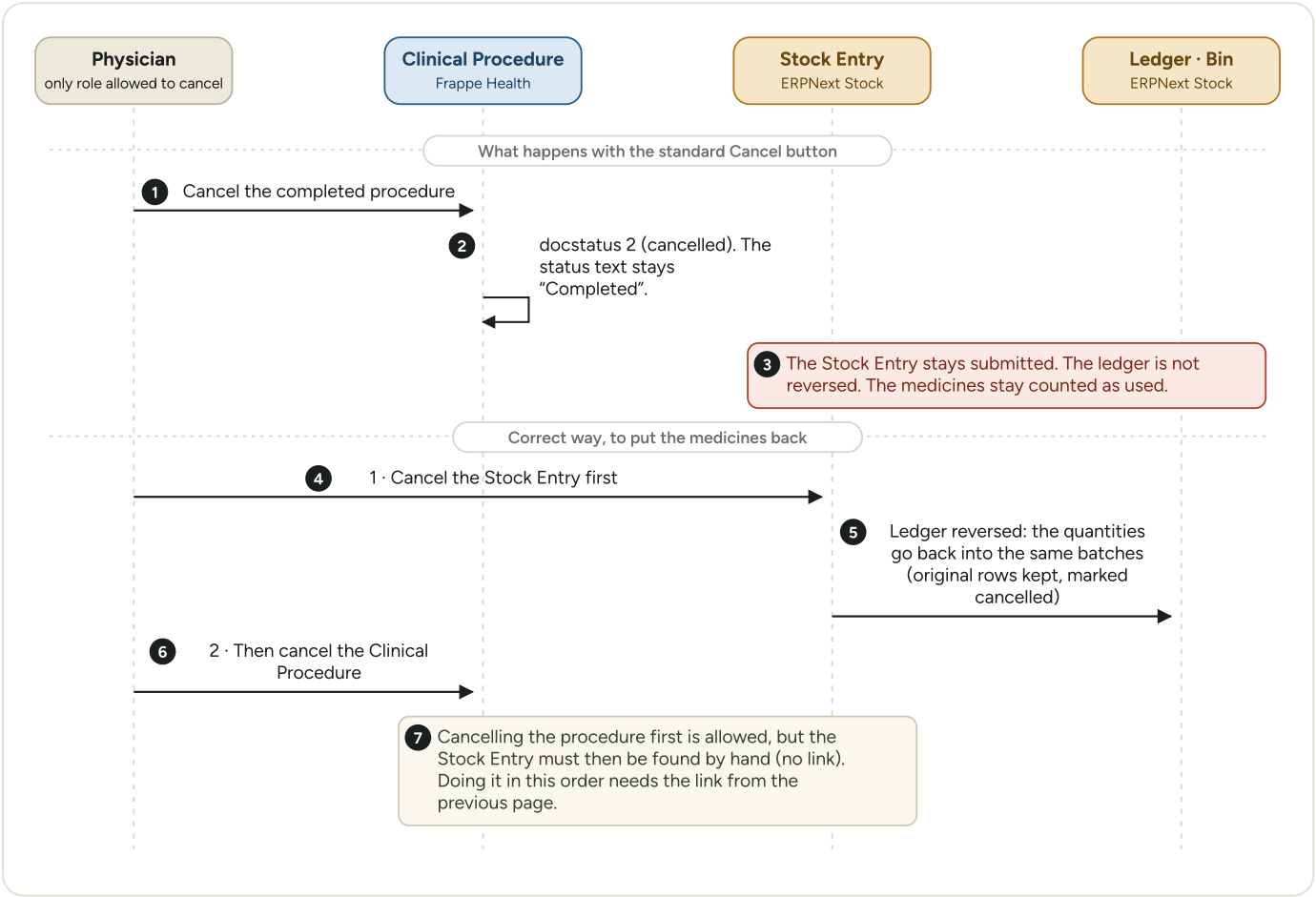
TEST	SETUP	RESULT
Earliest expiry first	Batch A expires in 100 days, batch B in 300 days; B was received first. Procedure uses 4.	**A** was used (the one expiring first)
Expired batch is skipped	Batch A made expired (6 left), B valid (10). Procedure uses 8.	All 8 came from **B**; A untouched
Not enough valid stock	B valid 10, A expired 10 (20 in the warehouse). Procedure uses 15.	Start says OK (it counts expired). **Complete is refused**: "Available qty 10 is less than Required Qty 15"
More than the whole warehouse	Procedure uses 25 with 20 in the warehouse.	Start answers "insufficient stock" and offers a Material Receipt
Item with no batches	Plain stock item, 50 received. Procedure uses 2.	Works; no batch involved
Receive into an expired batch	Material Receipt with an already expired batch.	Refused: "The batch … has already expired"

Reading the results. ERPNext protects expiry at the moment of posting: expired lots are never issued and valid lots are used oldest first. The Start check is only a first, rougher warning because it counts every unit in the warehouse.

CONNECTION 5

Cancelling a procedure

A procedure recorded for the wrong patient has to be undone. The standard Cancel does only half the job.



TESTED

- Cancelling a completed procedure (as Physician): the procedure becomes docstatus 2; its Stock Entry stays submitted; the warehouse quantity does not change.
- Nursing User cannot cancel procedures at all (only Physician can).

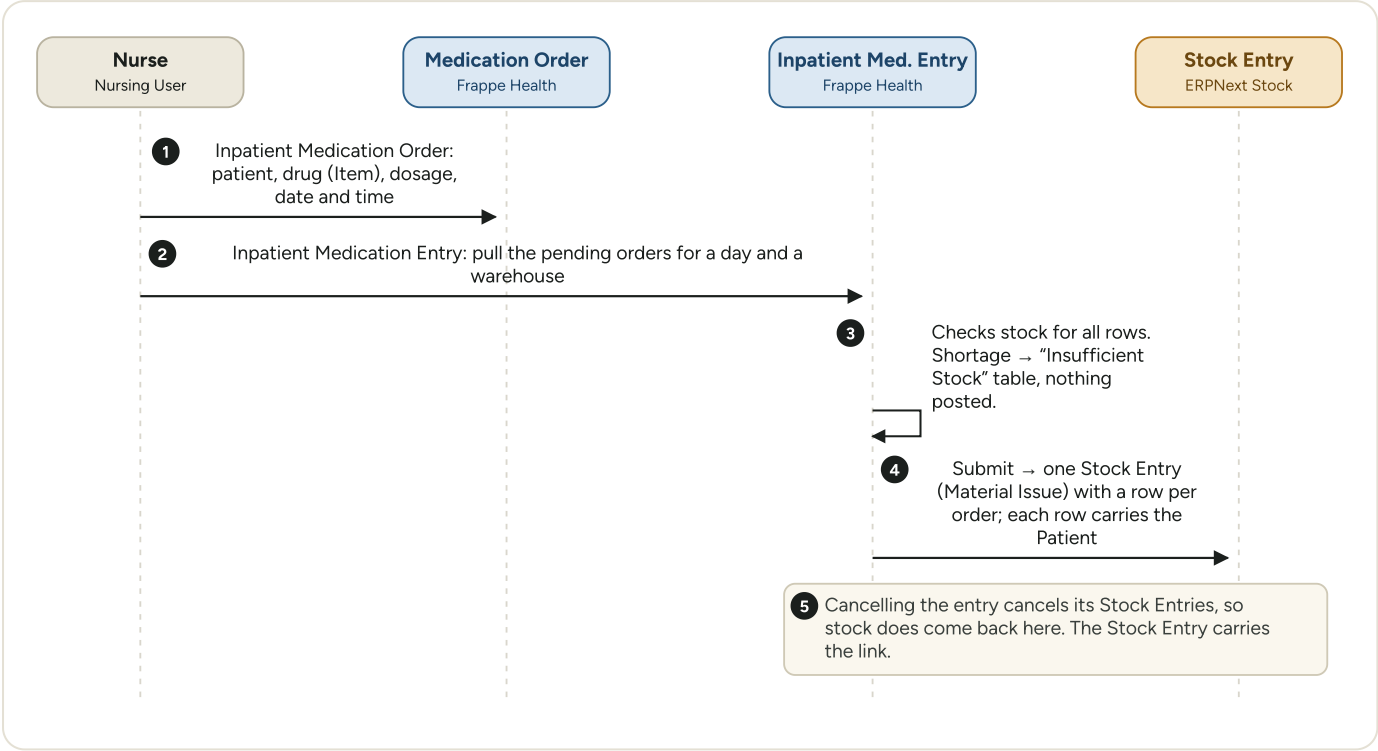
DECISION

- Either cancel the Stock Entry and the procedure together, in that order, every time.
- Or add a link from the procedure to its Stock Entry, so that one action cancels both.

CONNECTION 6

Inpatient medication

The second route from Health to Stock is for admitted patients. It works per order, not per procedure.



	CLINICAL PROCEDURE	INPATIENT MEDICATION ENTRY
Moves stock when	Complete and Consume	Submit
Patient on the Stock Entry	no	yes, on each row
Link to its Stock Entry	none	Stock Entry › Inpatient Medication Entry
Cancelling returns stock	no (standard)	yes
Used for	Care at the clinic: dressings, injections	Wards: scheduled doses for admitted patients

ALL ROUTES

Every way stock comes in and goes out

Stock only moves through a Stock Entry (or a document that creates one). These are all the routes, and the one we use for each situation.

STOCK IN

ROUTE	RECORD	WHEN	RECOMMENDED
Delivery with no purchase order	Stock Entry · Material Receipt	Donations, first stock, simple deliveries	**yes, default**
Delivery against a purchase order	Purchase Receipt	When purchasing is run in ERPNext	later
Move between warehouses	Stock Entry · Material Transfer	Store to ward	if needed
Correct a count	Stock Reconciliation	Annual or monthly count	yes, for counts
Return of a cancelled use	Cancel the Stock Entry	Wrong patient	see Connection 5

STOCK OUT

ROUTE	RECORD	WHEN	RECOMMENDED
Care given to a patient	Clinical Procedure → Material Issue	Dressing, injection, any care	**yes, default**
Admitted patient doses	Inpatient Medication Entry → Material Issue	Wards	if wards exist
Damaged, lost, thrown away	Stock Entry · Material Issue	Not tied to a patient	yes
Expired stock	Stock Entry · Material Issue (expired batch)	Clearing expired lots	yes
Sold or dispensed with an invoice	Sales Invoice with Update Stock, or Delivery Note	A pharmacy that bills each sale	if billing is needed
Count surplus or shortage	Stock Reconciliation	After a count	yes, for counts

SECURITY

Permissions in the modules

Standard roles, tested by creating users with one role each and trying each step. Cell = what that role can do on that record.

RECORD	NURSING USER	PHYSICIAN	HEALTHCARE ADMINISTRATOR	STOCK USER	ITEM MANAGER
Patient	read, write, create	read, write, create	—	**no access**	—
Medication	**no access**	**cannot create**	read, write, create	—	—
Procedure Template	read	read, write	read, write, create	—	—
Clinical Procedure	read, write, create, submit	also cancel	—	**no access**	—
Item	read	read	—	read	read, write, create
Stock Entry	**none**	—	—	read, write, create, submit, cancel	—
Material Request	—	—	—	read, write, create, submit	—

TESTED FINDINGS

- A **Nursing User alone cannot complete** a procedure that uses stock: creating the Stock Entry is refused. Adding the role **Stock User** fixes it; nothing else is needed.
- A Nursing User can create, submit and start a procedure, and read items and patients.
- A Stock User can receive stock and raise requests but cannot read patients or procedures.
- Only Physician can cancel a procedure.

A WORKABLE ROLE SET

- **Pharmacist:** Stock User + Item Manager (create medicines and batches).
- **Nurse:** Nursing User + Stock User (records and completes procedures).
- **Doctor:** Physician + Stock User if they complete procedures.
- **Administrator:** Healthcare Administrator (Medications, templates) + Stock Manager.
- **Site owner:** System Manager.

SETUP

Module setup checklist

What has to exist in the two modules before the flows above work. Nothing here needs custom code.

WHERE	SETTING	WHY
Company	Created by the setup wizard; if perpetual inventory is on, a Default Inventory Account	Stock Entries post to accounts
Warehouse	One warehouse for the pharmacy (e.g. “Pharmacy”)	Source of every issue, target of every receipt
Stock Settings	Pick Serial / Batch Based On = Expiry · Use Serial / Batch fields = on · Raise Material Request when stock reaches re-order level = on	Earliest expiry first, expired skipped, reorder
Healthcare Service Unit	Set its warehouse to the pharmacy warehouse (or choose the warehouse on each procedure)	Procedure warehouse default
Item defaults	Maintain stock, Has Batch No, Has Expiry Date, a batch series	Items made from Medications become batch and expiry items
Item Groups	A “Medicine” group with categories (analgesic, antibiotic…)	Lists, reports and medication classes
Healthcare Settings	Link Customer to Patient · patient naming · income / expense accounts	Patient companions, Stock Entry expense account
Price List	A default selling price list	Medications and templates need a price
Roles	See the role set above	Who may do what

Order matters. Create the Item defaults and Stock Settings first, then the Medications. Items already created keep their old settings.

DECISIONS

Gaps in the standard flow

These are properties of the standard modules, found by testing. Each one needs a decision, or a small addition.

GAP IN THE STANDARD MODULES	CONSEQUENCE	PROPOSED ANSWER
The procedure does not keep its Stock Entry number	No way to see which stock a procedure used, or to cancel the right entry	Add a link field on the Stock Entry to the Clinical Procedure, filled when completing
Standard Cancel does not return stock	A mistake leaves the medicines counted as used	Cancel the Stock Entry with the procedure, in that order, and keep a reason
Items from Medication are disabled and not batch / expiry items	Medicines cannot be used until fixed by hand	Item defaults for batch and expiry; enable the Item after creating the Medication
Start counts expired stock	A procedure can start with no valid stock and fail at Complete	Accept: the refusal at Complete protects stock; show valid stock to the nurse
A Stock Entry has no reason or recipient	“Why did stock leave?” is not recorded	Add two fields on the Stock Entry: reason and recipient
Nursing User cannot complete a stock-consuming procedure alone	Nurses are blocked at the last step	Give nurses Stock User as well
Nurses cannot cancel; only Physician	A wrong entry needs a doctor	Decide who may cancel; add the role or a Healthcare Administrator step
Prescriptions never move stock	Dispensing a prescription is not a stock movement	Dispense through a procedure, or bill with a Sales Invoice that updates stock

How we answer them. A link field, a reason field and a recipient field are added to the Stock Entry; Item defaults and Stock Settings are configured; the roles above are used. Everything else is standard.